

Krishna Srivastava

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Education

Heritage Institute of Technology, Kolkata

Aug 2023 – May 2027

B.Tech in Computer Science | SGPA: 8.0

Experience

Software & IT Intern

June 2025 – July 2025

Tata Power | On-site

- **Industrial Telemetry Dashboard:** Built a real-time monitoring dashboard to track the uptime and network status of IP cameras across the enterprise infrastructure, improving system reliability and industrial safety.
- **Automated Alerting Engine:** Engineered an automated Node.js WebSocket "heartbeat" service that actively detects client-side feed disconnections and triggers instant incident response workflows.
- **Backend & SQL Optimization:** Wrote and optimized advanced PostgreSQL queries to manage internal data, improving database retrieval performance for continuous telemetry streams.

Projects

AeroPredict: Aerospace Predictive Maintenance Software

Live | GitHub

Duration: April 2026 | Software Engineer

- Architected an end-to-end software pipeline utilizing the NASA C-MAPSS dataset to predict turbofan equipment failure, demonstrating technical approaches to reduce operational downtime in aerospace technologies.
- Developed a robust, decoupled backend using **Python FastAPI** to handle data preprocessing, creating an API layer that serves actionable insights directly to a React-based frontend dashboard.
- Engineered a machine learning module utilizing K-Means clustering and XGBoost to normalize multi-channel sensor feeds, applying strong problem-solving skills to capture complex equipment degradation trends.

Chronos: Distributed Microservice Architecture

GitHub

Duration: March 2026 – Present | Backend Systems Engineer

- Architected a fault-tolerant, highly concurrent microservice infrastructure utilizing **Python asyncio** and **ZeroMQ** (TCP) to orchestrate 100+ concurrent automation agents, bypassing OS-level thread bottlenecks.
- Engineered an automated data ingestion pipeline to extract and parse live external data, implementing a robust Fallback Matrix to gracefully handle API rate limits and network drops.
- Designed a high-throughput backend capable of structuring messy JSON payloads and processing 1,200+ network round-trips per second with strict polling timeouts for 100% reliability.

AERA: Automated Telemetry Orchestrator

Live | GitHub

Duration: March 2026

- Built an automated backend orchestrator using **Python FastAPI** to continuously query, extract, and parse massive volumes of complex spatial telemetry from external APIs (Open-Meteo, Overpass).
- Structured unstructured public web data into deterministic JSON blocks, creating clean data pipelines that feed directly into a containerized analysis engine.
- Containerized the application utilizing **Docker** and established secure deployment pipelines, ensuring high availability and HTTPS routing for dynamic frontend requests.

Skills

Programming Languages: Python, Java, C++ , C, JavaScript, TypeScript, SQL

Software Engineering Principles: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), System Design, Code Review Practices, Microservices Architecture

Backend & Systems: Node.js, FastAPI, RESTful APIs, ZeroMQ, WebSockets, TCP/IP

Data & Infrastructure: Docker, PostgreSQL, MongoDB, Git/GitHub, Linux, Applied AI/ML Integration

Positions/Achievements

- **Hackathon Leadership & Cross-Functional Collaboration:** Led diverse teams to secure a **Top 30 finish (out of 450+ participants)** at Hack Among Us (2026) by architecting a scalable backend platform, and a **Top 50 finish** at HackHeritage 3.0 (2025) for a sustainability project. Consistently demonstrated strong problem-solving skills under strict time constraints.
- **Member, Google Developer Group (GDG) HITK:** Active participant in technical code reviews, workshops, and community-driven collaborative software engineering initiatives.
- **Campus Ambassador, Techniche (IIT Guwahati):** Spearheaded campus outreach by organizing hands-on technical workshops and coding seminars, driving record student participation.